

# End-to-End Single Image Fog Removal Using Enhanced Cycle Consistent Adversarial Networks

Wei Liu<sup>ID</sup>, Xianxu Hou<sup>ID</sup>, Jiang Duan, and Guoping Qiu<sup>ID</sup>

**Abstract**—Single image defogging is a classical and challenging problem in computer vision. Existing methods towards this problem mainly include handcrafted priors based methods that rely on the use of the atmospheric degradation model and learning-based approaches that require paired fog-fogfree training example images. In practice, however, prior-based methods are prone to failure due to their own limitations and paired training data are extremely difficult to acquire. Moreover, there are few studies on the unpaired trainable defogging network in this field. Thus, inspired by the principle of CycleGAN network, we have developed an end-to-end learning system that uses unpaired fog and fogfree training images, adversarial discriminators and cycle consistency losses to automatically construct a fog removal system. Similar to CycleGAN, our system has two transformation paths; one maps fog images to a fogfree image domain and the other maps fogfree images to a fog image domain. Instead of one stage mapping, our system uses a two stage mapping strategy in each transformation path to enhance the effectiveness of fog removal. Furthermore, we make explicit use of prior knowledge in the networks by embedding the atmospheric degradation principle and a sky prior for mapping fogfree images to the fog images domain. In addition, we also contribute the first real world nature fog-fogfree image dataset for defogging research. Our multiple real fog images dataset (MRFID) contains images of 200 natural outdoor scenes. For each scene, there is one clear image and corresponding four foggy images of different fog densities manually selected from a sequence of images taken by a fixed camera over the course of one year. Qualitative and quantitative comparison against several state-of-the-art methods on both synthetic and real world images demonstrate that our approach is effective and performs favorably for recovering a clear image from a foggy image.

**Index Terms**—Single image defogging, cycleGAN, unpaired training, image restoration.

## I. INTRODUCTION

**F**OG is an atmospheric phenomenon caused by very small particles in the air that obscure the clarity of the atmosphere. In computer vision, fog can cause serious degradation in image quality which in turn can affect the performances of image analysis algorithms. Fog removal from a single image is a challenging problem that aims to recover a clean image from a given foggy image. Many defogging techniques in the literature are based on the atmospheric degradation model [26], [28], [29] which can be formulated as:

$$I(x) = J(x)T(x) + A[1 - T(x)] \quad (1)$$

where  $J(x)$  denotes the recovered image,  $I(x)$  is the observed hazy image, while the transmission map is denoted by  $T(x)$ , and  $A$  corresponds to the atmospheric light. Due to the unknown parameters  $A$ ,  $T$  and  $J$ , single image defogging is a mathematically ill-posed problem when we want to recover  $J$  from  $I$ . Thus, most single image defogging methods try to estimate the atmospheric light  $A$  and transmission  $T$  in order to recover a foggy image by using this model.

There are mainly two classes of defogging methods, prior-based and learning-based. Prior-based methods [1], [3], [9], [15], [46] obtain fog-related features through observation and statistics. It can be also called hand-crafted priors. However, these methods are usually under strict constraint conditions which can result in undesired fog artifacts in the recovered images. For example, He *et.al.* [15] assume that in a clear natural image, at least one channel in the RGB space is close to zero. This method may fail when dealing with scene objects which are similar to atmospheric light, such as sky or white building. Recently, deep learning-based methods [4], [20], [31], [41] are proposed to address the disadvantages of methods based on hand-crafted priors. They exploit the Convolutional Neural Network (CNN) to estimate the transmission and atmospheric light. However, these methods still use the defogging model (1) to recover the clear images. When the parameters of the model are not estimated accurately, defogging can introduce artifacts such as color distortion and halo. Furthermore, these methods require fog-fogfree image pairs to train their networks. Thus, they have to use the model in Equation (1) and labeled depth maps such as those from the NYU depth dataset [33] to synthesize fog-fogfree training

Manuscript received August 16, 2019; revised December 2, 2019 and April 29, 2020; accepted June 29, 2020. Date of publication July 13, 2020; date of current version July 17, 2020. This work was supported in part by the Education Department of Guangdong Province, China, under Project 2019KZDZX1028. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Chandra Sekhar Seelamantula. (Corresponding author: Guoping Qiu.)

Wei Liu and Xianxu Hou are with the Guangdong Key Laboratory for Intelligent Information Processing, Shenzhen University, Shenzhen 518060, China, also with the Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen University, Shenzhen 518060, China, also with the College of Electronics and Information Engineering, Shenzhen University, Shenzhen 518060, China, and also with the Shenzhen Institute of Artificial Intelligence and Robotics for Society, Shenzhen 518060, China.

Jiang Duan is with the School of Economic Information Engineering, SWUFE, Chengdu 610074, China.

Guoping Qiu is the Guangdong Key Laboratory for Intelligent Information Processing, Shenzhen University, Shenzhen 518060, China, also with the Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen University, Shenzhen 518060, China, also with the College of Electronics and Information Engineering, Shenzhen University, Shenzhen 518060, China, also with the Shenzhen Institute of Artificial Intelligence and Robotics for Society, Shenzhen 518060, China, and also with the School of Computer Science, University of Nottingham, Nottingham NG7 2RD, U.K. (e-mail: guoping.qiu@nottingham.ac.uk).

Digital Object Identifier 10.1109/TIP.2020.3007844

1057-7149 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

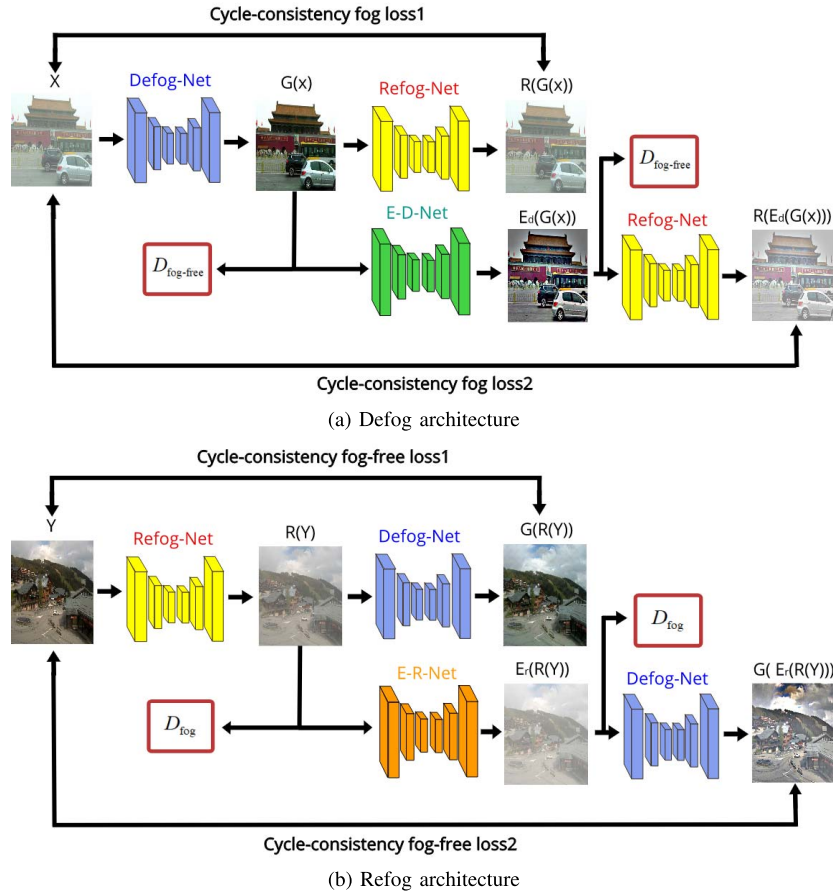


Fig. 1. The architectures of Cycle-Defog2Refog network.  $X$  is the input foggy image.  $G$  denotes the generator Defog-Net, and  $G(X)$  is the defogged image.  $Y$  is the clear image.  $R$  denotes the generator Refog-Net, and  $R(Y)$  is the synthetic foggy image.  $E_d$  denotes the generator Enhancer-Defog-Net (E-D-Net), which enhances the defogged image.  $E_r$  denotes the generator Enhancer-Refog-Net (E-R-Net), which enhances the synthetic image.  $D_{fog}$  is the adversarial discriminator for classifying the real foggy image and the generated foggy image.  $D_{fog-free}$  is the adversarial discriminator for distinguishing the real fog-free image and the defogged image.

image pairs. Synthesizing the data requires the depth maps of the fogfree images which are not always available, particularly for the outdoor scenes. For example, the NYU data [33] often used in the defogging literature is an indoor dataset and models trained with indoor data are not best suited for outdoor scenes.

In this paper, we present a novel fog removal system based on recent developments in adversarial generative network (GAN). Specifically, inspired by the principle of CycleGAN [45], we have developed an end-to-end learning system that uses unpaired fog and fogfree training images, adversarial discriminators and cycle consistency losses to automatically construct a defogging system. Similar to CycleGAN, our system has two transformation paths, one maps fog images to a fogfree image domain and the other maps fogfree images to a fog image domain. Instead of one stage mapping, our system uses a two stage mapping strategy in each transformation path (as shown in Fig. 1). In the fog to fogfree transformation path, fog images are mapped to a first fogfree domain. As the output of the first stage mapping may still contain residual fog, they are passed onto a second mapping network to remove the fog further. Similarly, in the fogfree to fog transformation path, fogfree images are transformed to a fog image domain first

and the results are then passed onto a second transformation network to the fog image domain. In constructing the fogfree to fog domain transformation, we explicitly embed the atmospheric degradation model (1) in the learning process. Furthermore, a sky prior is introduced to reduce artifacts. We present experimental results on synthetic foggy and natural foggy image datasets to show the effectiveness of the new defogging technique.

To our knowledge, there is currently no real world fog-fogfree nature image dataset suitable for defogging research publicly available. We believe using clear and foggy images that occur naturally is valuable for developing practically useful defogging technologies. In this work, we have collected clear and foggy images of 200 natural outdoor scenes. Each scene was imaged by a fixed camera over the course of one year. One clear image and four foggy images of different fog intensities were manually selected to form the Multiple Real Fog Images Dataset (MRFID). We believe this dataset will be useful for the research community and we will make this dataset publicly available to researchers to facilitate the development of defogging techniques. The dataset will be made available online in due course.

The organization of the paper is as follows. In Section 2, we briefly review relevant literature. In Section 3, we present the new cycle fog to fogfree learning framework, its components, learning cost functions and training procedure. Section 4 presents experimental results and Section 5 concludes the paper.

## II. RELATED WORK

In this section, we mainly review relevant single image defogging methods, which can be roughly grouped as prior-based methods and learning-based methods.

### A. Prior-Based Methods

These methods have been widely used in the past few years and are also known as hand-crafted feature based methods. These methods often leverage the statistics of the natural image to characterize the transmission map, such as Dark-channel Prior [15], Color Attenuation Prior [46], Contrast Color-lines [9], Hue Disparity Prior [1], Haze-line Prior [3], Configurable Contrast Enhancement Model [24], and Multi-scale Optimal Fusion dehazing method [44]. Particularly, the method of the dark-channel has shown its excellent defogging performance, which has led many researchers to improve this method to achieve single image defogging. Despite the remarkable defogging performance by these methods, hand-crafted features (such as textural, contrast and so on.) also have limitations. For instance, dark-channel prior [15] does not work well for some scene objects (such as sky, white building and so on.) which are inherently similar to the atmospheric light. Using haze-line prior [3] can cause color distortion when the fog density is high.

### B. Learning-Based Methods

Recently, some learning-based methods have drawn significant attention in the defogging research community. Tang *et.al.* [37] proposed a method by using the random forest methods to train dark primary colors and other multiple color features to improve the estimation accuracy of transmittance. Mai *et.al.* [25] found that the RGB color feature of the haze image had a strong linear relationship with the depth of the scene, and established the intrinsic relation between the color feature and the scene depth through the back-propagation neural network to effectively restore the scene depth. Cai *et.al.* [4] proposed a concept of a dehazing network that used a convolutional neural network to train the color characteristics (such as dark primary colors, color fading, maximum contrast, etc.) of foggy images and to optimize the transmission. All of those methods can achieve a better defogging effect. However, they still have to estimate the transmission map and atmospheric light first, and then remove the fog with the atmospheric degradation model. Thus, the artifacts could not be avoided in the final defogged results when the transmission or atmospheric light is wrongly estimated.

To address the above problem, networks based on encoder-decoder structure have been used to directly recover clear images. Zhang *et.al.* [42] introduced a multi-scale image

dehazing method using the Perceptual Pyramid Deep network based on the dense blocks and residual blocks to directly remove the fog from a single foggy image. However, this method used fog-fogfree training image pairs. Wang *et.al.* [38] observe that the atmospheric illumination in hazy weather mainly has a great influence on the luminance channel in YCrCb colorspace and has less impact on the chrominance channels. Thus, they use this prior with multiscale convolutional networks to recover the image. There is the same problem as Zhang's method [42], which needs paired training data. Yang *et.al.* [40] proposed a proximal Dehaze-Net to learn the dark channel and transmission priors and achieved a good defogging performance. Different from the above training mode, this method requires transmission maps and dark channels in addition to the paired training data. Based on a fusion strategy, Ren *et.al.* [32] introduced an end-to-end trainable neural network to obtain three haze-relevant confidence maps (White Balance, Contrast Enhancing, Gamma Correction). Finally, they gate the important features of these maps to yield the defogged image. This method can recover the foggy images well, however, there is still some fog remain in the defogged results. The reason is that although these confidence maps can be used to represent the haze-relevant features, they do not reflect the depth of the scene very well. Moreover, this method also needs paired fog-fogfree data to train the network. Different from Ren's fusion strategy, Akshay *et.al.* [7] first proposed two convolutional neural networks to generate two transmission maps from a foggy image through RGB and YCbCr color spaces respectively. And then they introduced a transmission fusion network to integrate these two transmission maps to obtain a robust transmission map for a foggy scene. However, this method still needs the model (1) to recover the fog-free scene and the paired training data is required too.

Except for the above methods, generative adversarial network (GAN) [12] based defogging algorithms have achieved remarkable results. Li *et.al.* [22] modify the basic GAN to directly restore a clear image from a foggy image. However, this method also requires fog-fogfree pair images to train the network. In practice, it is difficult to obtain a large number of paired fog-fogfree images. A method based on CycleGAN [45] has been proposed in [8] where cycle-consistency and VGG perceptual losses are used to directly remove fog. A significant advantage of using CycleGAN is that there is no need to use paired fog-fogfree images to train the system. However, due to the one stage mapping strategy, this method is liable to cause color distortion, low contrast and loss of texture information in the image after defogging. Thus, inspired by the advantage of this adversarial learning approach and to overcome its defect, we show, for the first time, that a two stage mapping strategy based on cycle consistent adversarial networks with prior knowledge has a remarkable performance for image defogging. Besides, the proposed method fills a gap in the literature of image defogging by introducing a new method that not only learns from non-pairing training samples but also explicitly incorporates the atmospheric degradation principle and a sky prior in the learning model.



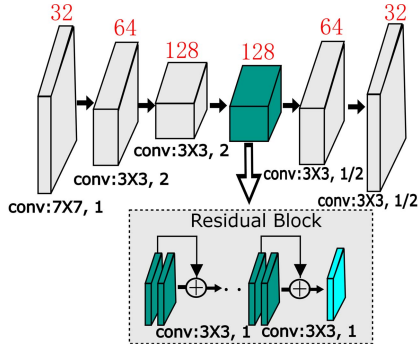


Fig. 2. The defog network architecture. Here, (Conv:3 × 3, 1/2) denotes a convolutional layer with kernel size of 3 and stride of 1/2.

### III. CYCLE-DEFOG2REFOG NETWORK

In this section, we introduce the details of the proposed network. It consists of two parts, a defog architecture and a refog architecture, as shown in Fig. 1. In the defog architecture, we use a refog-net ( $R$ ) and an enhancer-defog-net ( $E_d$ ) to constrain the defogging mapping function with two consistency fog loss functions and an adversarial discriminator  $D_{fog-free}$ , as shown in Fig. 1 (a). In the refog architecture, a defog-net ( $G$ ) and an enhancer-refog-net ( $E_r$ ) are used to supervise the refogging mapping function with two consistency fog-free loss functions and an adversarial discriminator  $D_{fog}$ , as shown in Fig. 1 (b).

#### A. Defog-Net

The details of the defog network are shown in Fig. 2. For the generator  $G$ , we adopt an encoder-decoder network from Johnson [18]. We use 3 convolutional blocks in the encoding process and 3 deconvolutional blocks in the decoding process. In the encoder network, the first layer has 32 filters with a kernel size  $7 \times 7$  and a stride of 1, the second layer has 64 filters with a kernel size of  $3 \times 3$  and a stride of 2, and the third layer has 128 filters with a kernel size of  $3 \times 3$  and a stride of 2. Each layer of the decoder network has the same number of filters as its symmetric layer in the encoder but their convolutions have a stride of 1/2. Moreover, similar to CycleGAN, we have used 9 residual blocks [16].

#### B. Refog-Net

In the traditional CycleGAN, we can use a CNN to directly generate the foggy images to constrain the defog mapping function. However, in practice, it is extremely difficult to fit the distribution of the foggy image by only using a convolutional neural network due to the diversity and complexity of a variety of foggy image contents. Instead, we introduce a CNN based atmospheric degradation model to synthesize foggy images. Specifically, we use a CNN to estimate the transmission map  $T$  and use a sky prior to estimate the atmospheric light  $A$ . In Section V-A, we qualitatively and quantitatively discuss the differences between the generated results by using our proposed method and CycleGAN in detail.

The details of the refog network are shown in Fig. 3. We introduce a CNN to estimate the transmission map  $T$ . Each

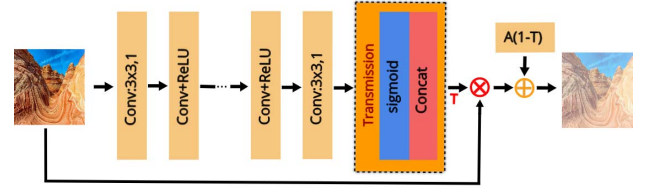


Fig. 3. The details of the refog network. Here, (Conv:3 × 3, 1) denotes a convolutional layer with kernel size of 3 and stride of 1.  $T$  denotes the transmission map.  $A$  denotes the atmospheric light.

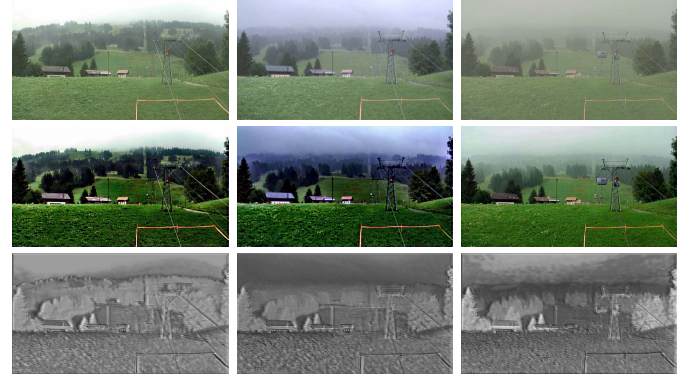


Fig. 4. Defogging results of a scene with different fog densities from MRFID. Top row: the same scene with different fog densities. Middle row: corresponding defogging images by our Cycle-Defog2Refog method. Bottom row: corresponding transmission maps.

layer of this network has 64 filters with a kernel size of  $3 \times 3$  and a stride of 1. In our network, we use 5 layers to estimate the transmission  $T$ . As shown in Fig.4, three transmission maps of the same scene with different fog densities are generated through this network. We can see that, in different fog densities, our network can capture the edge information and outline the profile of the objects.

In addition to  $T$ , we need to estimate  $A$  in order to use the atmospheric degradation model (1). It is well known that the more accurately  $A$  is estimated, the better a defogging performance can be obtained. He *et.al.* [15] first selected the top 0.1 percent brightest pixels in the dark channel of the foggy image and then took the maximum of these top pixels as  $A$ . Tan *et.al.* [36] picked the highest intensity of foggy image as the atmospheric light value. However, taking the maximum value of the pixels as  $A$  is prone to color distortion or halo artifact in the defogged results. To address this problem, we first segment the sky region from the foggy images and then take the average pixel values of the sky region as  $A$ . The depth of the sky in an image is regarded as infinity, i.e.,

$$d_{sky}(x) \rightarrow +\infty \quad (2)$$

In model (1), the transmission  $T(x)$  is usually described as the portion of the light which reaches the camera from the object and can be expressed as follows [26]:

$$T(x) = e^{-\beta d(x)} \quad (3)$$

where  $\beta$  is the scattering coefficient of the atmosphere and it is always assumed constant.  $d(x)$  is the depth from the scene point to the camera. Substitute equation (2) into (3)

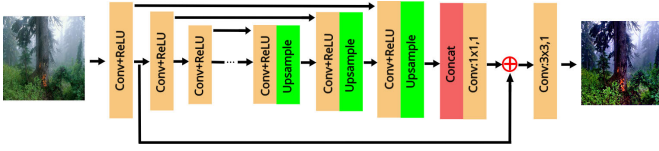


Fig. 5. The architecture of E-Net.

and calculate the limit of both sides:

$$T_{sky}(x) \approx 0 \quad (4)$$

Substitute equation (4) into (1):

$$I(x) \approx A \quad (5)$$

The value of  $A$  in equation (5) can be considered as the intensity of pixels in the area of maximum fog density [36]. Thus, it is reasonable to take the average value of the sky region as  $A$ . Then, the atmospheric light can be calculated as:

$$A_{sky} = \text{mean} (I_{sky}^c(x))_{c \in \{r, g, b\}} \quad (6)$$

where  $I_{sky}^c$  is a color channel of the sky region in a foggy image,  $\text{mean}$  is an average filter to process each pixel  $c \in \{r, g, b\}$  for each RGB channel. Moreover, for the sky segmentation algorithm, we can choose the method of image matting [14], [19], [35] or OSTU [30]. For a foggy image with no sky region or few sky region, we take the atmospheric light value  $A$  follows the method of He [15]. We picked the top 0.1 percent brightest pixels in the dark channel and took the maximum of these pixels corresponding to the original image as the value of  $A$ . As shown in Fig. 3, when the  $T$  and  $A$  are obtained, we synthesize the foggy image by using atmospheric degradation model (1).

### C. E-Net

Although the sky prior strategy can prevent the artifacts, the defogged images generally have low contrast and can lose some of the texture information by the remaining fog. As shown in Fig. 6 (b), the whole image looks dim and the details of edges are not clear. To overcome this shortcoming, we introduce an enhancer network to improve the quality of the generated images. It is another important new feature in our approach. We refer to this network as E-Net, and its architecture is shown in Fig. 5.

In E-Net, we also use the encoder-decoder structure. In the encoder network, we used 3 convolutional blocks, each layer is of the same type: 64 filters of a kernel size  $3 \times 3$  with stride 2. Different from the encoder, the decoder network has 5 deconvolutional blocks, each layer of the first three blocks has the same type: 64 filters of a kernel size  $3 \times 3$  with stride 1. Note that, in each deconvolutional block, there are several skip connections to be added by an element-wise summation from the convolutional blocks, which enforces the network to learn more details. In the fourth deconvolutional block, it consists of a concatenation function and a convolutional layer with 64 filters of a kernel size  $1 \times 1$ . Then, we multiply the output of the first convolutional blocks and the fourth deconvolutional blocks as the input to the last layer.

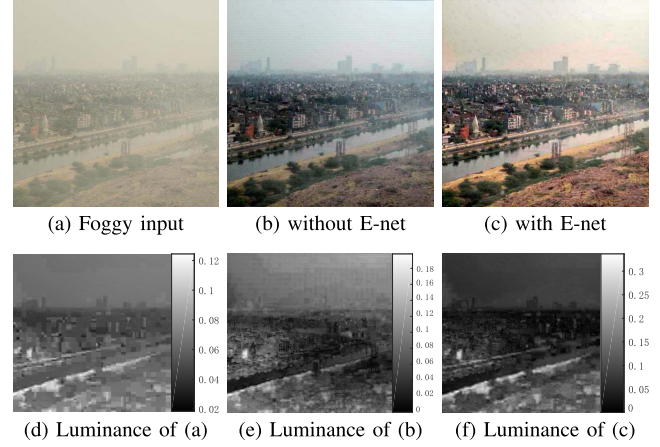


Fig. 6. Defogged results by using the Cycle-Defog2Refog without/with E-net and the corresponding luminance weight map.

The last deconvolutional layer is with a kernel size of  $3 \times 3$  and stride of 1.

In practice, this network plays the role of enhancer in Defog-net, while a ‘de-enhancer’ in Refog-net. Thus, as shown in Fig. 1, we refer it as Enhancer-Defog Net (E-D-Net,  $E_d$ ) in the Defog architecture, which enhances the image texture features of the generator  $G$ . In the Refog architecture, we refer it as Enhancer-Refog Net (E-R-Net,  $E_r$ ), which increases the fog density of the generator  $R$ . Fig. 6 shows an example that illustrates the advantage of including the E-Net in the system. We can see that the defogged result with the E-net looks clearer than the one without the E-net. Moreover, this is reflected in the Luminance weight map. This map is used to measure the visibility of each pixel in the image and assigns high values to areas with good visibility and low values to areas with poor visibility. As shown in Fig. 6 (f), the result with E-net has larger dynamic range of luminance values ( $0 \sim 0.3$ ) than the other in Fig. 6 (e) ( $0 \sim 0.18$ ). Due to the remaining fog in the result without E-net (Fig. 6(b)), the texture details in its luminance weight map are not clear (as shown in Fig. 6 (e)). In contrast, as shown in Fig. 6 (f), the texture details are clearer.

### D. Discriminator

The function of the discriminator is to distinguish whether an image is real or fake. In our network, we have two discriminators. As shown in Fig. 1,  $D_{fog}$  is used to distinguish between the input foggy images and the generated images from  $R$ ; in the same way,  $D_{fog-free}$  is used to discriminate between the generate images from  $G$  and the input clear images. For the discriminator networks, we use 5 convolutional blocks to classify whether the images are real or fake. Each layer has the same kernel size  $4 \times 4$  with a stride of 2, and the filters are 64, 128, 256, 512, 1 from lowest to highest.

### E. Full Objective of Cycle-Defog2Refog

In this work, we use the hybrid loss function. We first use the generative adversarial loss [12] to supervise the network. For the refog mapping function  $R$  and its discriminator  $D_{fog}$ ,

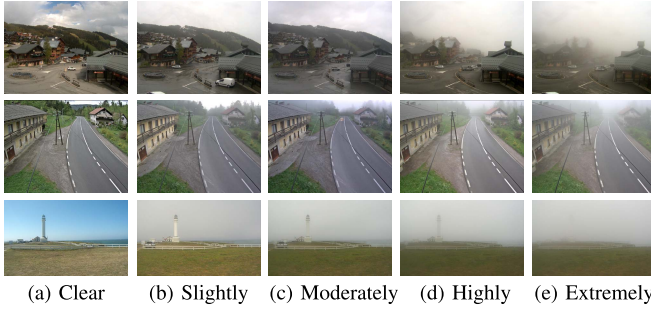


Fig. 7. The image samples from MRFID.

the adversarial loss of refof is defined as follows:

$$\begin{aligned} \mathcal{L}_{r-adv}(R, D_{fog}, Y, X) \\ = \mathbb{E}_{X \sim p_{fog}(X)} [\log D_{fog}(X)] \\ + \mathbb{E}_{Y \sim p_{clear}(Y)} [\log(1 - D_{fog}(R(Y)))] \end{aligned} \quad (7)$$

where,  $R$  is our refof network in Fig. 1, while  $D_{fog}$  aims to distinguish between the generated foggy images by  $R(Y)$  and the real foggy images  $X$ . In a similar way, the defog adversarial loss is defined as follows:

$$\begin{aligned} \mathcal{L}_{d-adv}(G, D_{fog-free}, X, Y) \\ = \mathbb{E}_{Y \sim p_{clear}(Y)} [\log D_{fog-free}(Y)] \\ + \mathbb{E}_{X \sim p_{fog}(X)} [\log(1 - D_{fog-free}(G(X)))] \end{aligned} \quad (8)$$

In the above equation,  $G$  and  $R$  are able to minimize the objective against the adversary  $D_{fog}$  and  $D_{fog-free}$  that try to maximize it. However, there will be some artifacts in the generated results when we only use the adversarial loss. We also found that it is difficult to remove the fog from foggy images by only using this loss function. The details are discussed later in Section 5.

Once the generated images  $G(X)$  and  $R(Y)$  are enhanced by E-Net, the loss function is defined as follows:

$$\begin{aligned} \mathcal{L}_{Enhancer} = \|E_d(X) - R(E_d(G(X)))\|_2^2 \\ + \|E_r(Y) - G(E_r(R(Y)))\|_2^2 \end{aligned} \quad (9)$$

where  $E_d$  denotes the E-D-Net,  $E_r$  denotes the E-R-Net.  $X$  is the foggy image,  $Y$  is the fog-free image.

In order to better remove fog and preserve more detail of texture information, we introduce another loss function, which is defined as Cycle-Refof loss. This function is used to minimize the objective between the foggy images  $X$  and its reconstructed foggy images  $R(G(X))$ , the clear images  $Y$  and its reconstructed clear images  $G(R(Y))$  (as shown in Fig. 1). We formulate this objective as:

$$\mathcal{L}_{Cycle-Refof} = \|X - R(G(X))\|_2^2 + \|Y - G(R(Y))\|_2^2 \quad (10)$$

Moreover, to learn more textural information from foggy images, a perceptual loss based on pre-trained VGG16 [34] is introduced to further constrain the generators, which is defined as:

$$\begin{aligned} \mathcal{L}_{VGG} = \|\mathcal{P}_i(X) - \mathcal{P}_i(R(G(X)))\|_2^2 \\ + \|\mathcal{P}_i(Y) - \mathcal{P}_i(G(R(Y)))\|_2^2 \end{aligned} \quad (11)$$

### Algorithm 1 The Training Procedure for Our Network

#### Input:

The foggy image set,  $X_n$ ;  
The clear image set,  $Y_n$ ;

#### Output:

The defogged image,  $I$ ;  
The refofged image,  $II$ ;

```

1: loop
2:   for  $x \in X_n, y \in Y_n$  do
3:      $I = f_{defog}(x) \leftarrow$  defog with Defog-Net;
4:      $I' = f_{refog}(I) \leftarrow$  refof with Refog-Net;
5:      $I'' = f_{enhancer}(I) \leftarrow$  enhance with E-D-Net;
6:      $I''' = f_{refog}(I'') \leftarrow$  refof with Refog-Net;
7:      $x' = f_{enhancer}(x) \leftarrow$  enhance with E-D-Net;
8:      $\mathcal{L}_f = \text{mean} \|x - I'\| \leftarrow$  fog consistent loss (Eq.10);
9:      $\mathcal{L}_{f-e} = \text{mean} \|x' - I'''\| \leftarrow$  fog-enhance consistent loss (Eq.9);
10:     $\mathcal{L}_{d-adv1} = \text{mean} \|1 - D_{fog-free}(y) + D_{fog-free}(I)\| \leftarrow$  defog adversarial loss1 (Eq.8);
11:     $\mathcal{L}_{d-adv2} = \text{mean} \|1 - D_{fog-free}(y) + D_{fog-free}(I'')\| \leftarrow$  defog adversarial loss2 (Eq.8);
12:     $\mathcal{L}_{d-v} = \text{mean}(VGG(x, y, f_{defog}, f_{refog})) \leftarrow$  defog VGG loss (Eq.11);
13:     $\mathcal{L}_{defog} = \mathcal{L}_f + \mathcal{L}_{f-e} + \mathcal{L}_{d-adv1} + \mathcal{L}_{d-adv2} + \mathcal{L}_{d-v}$ 
14:    return  $I$ 
15:   end for
16:   for  $y \in Y_n, x \in X_n$  do
17:      $II = f_{refog}(y) \leftarrow$  refof with Refog-Net;
18:      $II' = f_{defog}(II) \leftarrow$  defog with Defog-Net;
19:      $II'' = f_{enhancer}(II) \leftarrow$  enhance with E-R-Net;
20:      $II''' = f_{defog}(II'') \leftarrow$  defog with Defog-Net;
21:      $y' = f_{enhancer}(y) \leftarrow$  enhance with E-R-Net;
22:      $\mathcal{L}_{ff} = \text{mean} \|y - II'\| \leftarrow$  fog-free consistent loss (Eq.10);
23:      $\mathcal{L}_{ff-e} = \text{mean} \|y' - II'''\| \leftarrow$  fog-free enhance consistent loss (Eq.9);
24:      $\mathcal{L}_{r-adv1} = \text{mean} \|1 - D_{fog}(x) + D_{fog}(II)\| \leftarrow$  refof adversarial loss1 (Eq.7);
25:      $\mathcal{L}_{r-adv2} = \text{mean} \|1 - D_{fog}(x) + D_{fog}(II'')\| \leftarrow$  refof adversarial loss2 (Eq.7);
26:      $\mathcal{L}_{r-v} = \text{mean}(VGG(y, x, f_{defog}, f_{refog})) \leftarrow$  refof VGG loss (Eq.11);
27:      $\mathcal{L}_{refog} = \mathcal{L}_{ff} + \mathcal{L}_{ff-e} + \mathcal{L}_{r-adv1} + \mathcal{L}_{r-adv2} + \mathcal{L}_{r-v}$ 
28:     return  $II$ 
29:   end for
30: end loop

```

where  $\mathcal{P}_i$  denotes the feature maps of the  $i$ -th layer of the VGG16 network.

Finally, by combining the refof adversarial loss, defog adversarial loss, enhancer loss and Cycle-Refof loss, our overall loss function is:

$$\begin{aligned} \mathcal{L}_{Cycle-Defog2Refof} = \gamma_1 \mathcal{L}_{r-adv} + \gamma_2 \mathcal{L}_{d-adv} \\ + \gamma_3 \mathcal{L}_{Cycle-Refof} + \gamma_4 \mathcal{L}_{Enhancer} \\ + \gamma_5 \mathcal{L}_{VGG} \end{aligned} \quad (12)$$



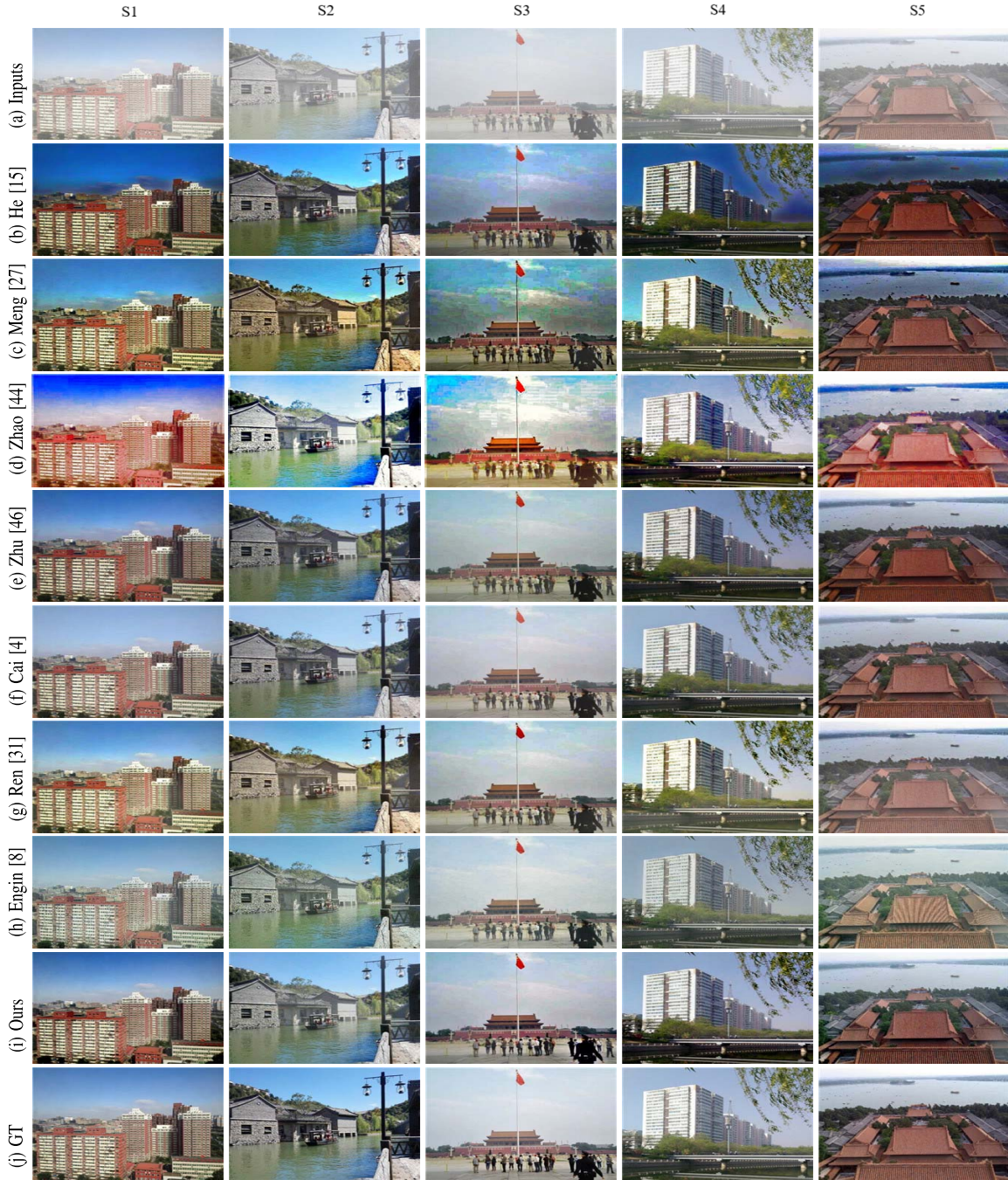


Fig. 8. Qualitative evaluation of defogged results on the synthetic images.

where  $\gamma_1, \gamma_2, \gamma_3, \gamma_4$  and  $\gamma_5$  are the positive weights. To optimize our network, we aim to solve:

$$D^*, R^* = \arg \min_{G, R, E} \max_{D_{fog}, D_{fog-free}} \mathcal{L}_{Cycle-Defog2Refog} \quad (13)$$

We use the generator  $D^*$  to remove the fog from the testing image. The overview of our training procedure for Cycle-Defog2Refog is shown in Algorithm 1.

#### IV. EXPERIMENTAL RESULTS

In this section, we qualitatively and quantitatively evaluate the defogged results by our proposed method against

seven state-of-the-art methods on synthetic and real-world images. The representative methods we compare our method against include Cai *et.al.* [4], Zhao *et.al.* [44], Engin *et.al.* [8], Ren *et.al.* [31], He *et.al.* [15], Zhu *et.al.* [46] and Meng *et.al.* [27]. Moreover, we use four evaluation criteria: Peak Signal-to-Noise Ratio (PSNR) [5], Structural Similarity (SSIM) [39], Fog Aware Density Evaluator (FADE) [6] and Blind Assessment based on Visibility Enhancement (BAVE) [13]. Among them, the first two are full reference image quality assessment(FRIQA) methods, and the last two are blind image quality assessment(BIQA) methods. Due to the fog-free reference image is difficult to obtain, it is a difficult





Fig. 9. Qualitative evaluation of defogged results on the real world images.

task to quantitatively evaluate the real-world defogged images. Thus, the FRIQA is not suitable for the evaluation of this task. To the best of our knowledge, BAVE and FADE are the few existing methods designed to quantitatively evaluate defogging performance. FADE (represented as  $\mathcal{F}$ ) predicts the visibility of a single foggy image to evaluate the fog density. The lower the value of the FADE, the better the defogged performance. In BAVE, there are three indicators to evaluate the defogged result. They are the rate of new visible edges  $e$ , the quality value of the contrast restoration  $\bar{r}$  and the normalized saturate value of pixels  $\delta$ . For a defogged result, the higher values of  $e$  and  $\bar{r}$ , and the smaller value of  $\delta$ , the better the recovered quality.

#### A. Datasets

Because our method does not need fog-fogfree image pairs to train the model, we do not have to synthesize the training samples, which is usually adopted by existing deep learning-based defogging methods. We collect the clear and foggy images from Google site, RESIDE dataset [21] and our own multiple real foggy images defogging (MRFID) datasets. MRFID contains images from 200 clear outdoor scenes. For each of the clear images, there are four images of the same scene containing different densities of fog defined as slightly foggy, moderately foggy, highly foggy and extremely foggy and some examples are shown in Fig. 7. The foggy images of



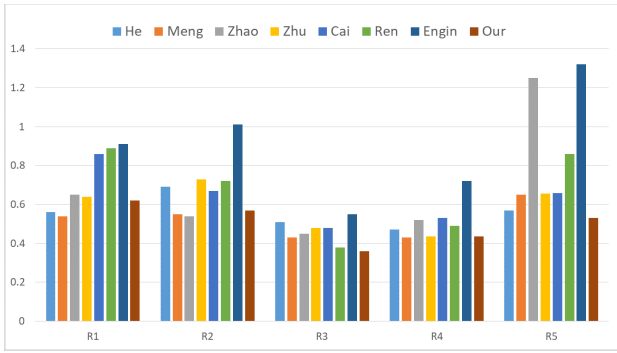


Fig. 10. Quantitative evaluate  $\mathcal{F}$  of the defog results in Fig. 9.

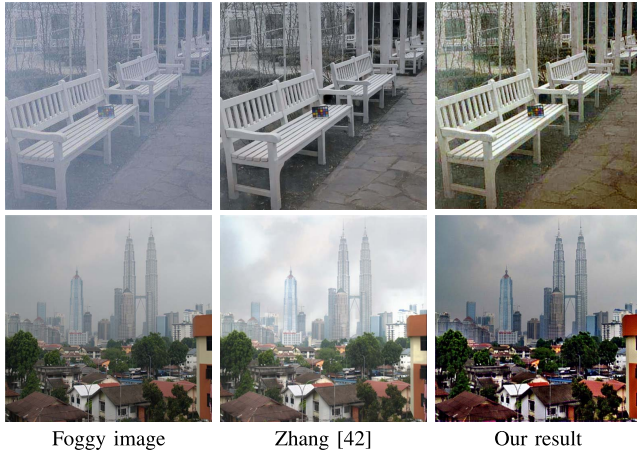


Fig. 11. Comparison of defogged images using Zhang [42] and Cycle-Defog2Refog.

MRFID are selected from an image dataset called the Archive of Many Outdoor Scene (AMOS) [17], in which images were captured by 29,945 static webcams located around the world, and contains 1,128,087,180 images from 2006 to 2017. In MRFID, images of each scene were manually selected from images taken within one year period, the image sizes range from  $640 \times 480$  to  $2,284 \times 914$ . Thus, this new foggy dataset will be useful for research on deep learning-based defogging methods. Moreover, in this paper, the number of training foggy images is 12,461, and 1,000 for testing. The number of training clear images is 12,257.

### B. Implementation Details

In our network, all the training samples are resized to  $512 \times 512$  for implementation convenience. For testing, the image is first resized to  $512 \times 512$  for defogging and then resize the defogged image back to its original size for computing the various performance measures. We empirically choose  $\gamma_1 = 10$ ,  $\gamma_2 = 10$ ,  $\gamma_3 = 8$ ,  $\gamma_4 = 5$  and  $\gamma_5 = 2$  for the loss function in generating the fog-free image. During training, ADAM optimizer is used for the generator and discriminator with the learning rate of  $2 \times 10^{-4}$  and batch size of 1. For all the defogged results in this paper, we used TensorFlow [10] to train the network for 35,400 iterations, which takes about 126 hours on an NVidia GeForce GTX 1080 Ti GPU.



Fig. 12. Comparison of defogged images using Wang [38] and Cycle-Defog2Refog.

### C. Qualitative and Quantitative Evaluation

For qualitative evaluation, we compare our proposed method against four prior-based and four deep learning-based state-of-the-art methods [4], [15], [27], [31], [38], [42], [44], [46] on both synthetic and real world images.

1) *Evaluation on Synthetic Images:* In this experiment, the synthetic images have come from RESIDE dataset [21]. Fig. 8 shows several defogged results by our method and other defog approaches. It can be observed that there is a phenomenon of color distortion in the sky region of He's, Meng's and Zhao's results (as can be seen in Fig. 8 (b), Fig. 8 (c) and Fig. 8(d)). The reason is due to these three prior-based methods overestimate the atmospheric light. Moreover, as observed in Fig. 8 (e)-(g), we note that our results are similar to the defogged results of Zhu's, Cai's and Ren's methods. Although there are no artifacts in the results by using learning-based methods [4], [31], some remaining fog is still not removed as shown in Fig. 8 (f)-(g). As shown in Fig. 8(h), all of the defogged images have a thin veil.

In comparison, our model can generate clearer images from the single foggy images and has no artifacts, as shown in Fig. 8 (i). This is because our enhancer network can improve the texture details of the outputs from generator  $G$  and  $R$ , which can further strengthen mapping capability and preserve the contrast and sharpness of the image. It is also reflected in the quantitative results in Table I. It can be seen that the PSNR in our results are larger than the other methods. Moreover, the SSIM in our results are always larger than others. It is demonstrated that our method has an overall better defog performance on synthetic images compared with other defog methods.

2) *Evaluation on Real World Images:* We evaluate our method on real world images which are provided by the authors of previous methods and available on the Internet [9], [32]. Fig. 9 shows the comparison of defogged results by our method and the other five defog methods. As shown in Fig. 9 (b)-(c), there are still some artifacts in He's and Meng's results, e.g. color distortion in the sky region in the R2 column. In addition, the defogged results by He's method [15] are darker than other methods due to overestimating the fog





Fig. 13. Qualitative comparison of different methods on the MRFID dataset.

density. There is an over enhancement phenomenon in Zhao's results [44], as shown in Fig. 9(d), e.g. the color of the people's clothes are over enhancement in R5 column. The results of Zhu [46] as shown in Fig. 9 (e) have a good performance, however, the defogged images are darker than ours in luminance. The results of Cai [4] and Ren [31] as shown in Fig. 9 (f)-(g) are similar. However, there are still some remaining fog which are not removed from the image. As shown in Fig. 9 (h), there is a visible fog in the defogged images by using Engin [8] method. In contrast, the results of our method as shown

in Fig. 9 (i) are more natural and clearer. It is also reflected in the quantitative results in Table II and Fig. 10. Regarding indicator  $\delta$ , its value always maintained in a small range, which means that our results have a good saturation. Regarding indicators  $e$  and  $\bar{r}$ , our results have higher values than others. It demonstrates that the local contrast and edge information are recovered well in our results. Moreover, as shown in Fig. 10, we can see that the FADE values of our results are almost smaller than other methods. It turns out that the texture details of our defogged results are recovered felicitously and



TABLE I  
QUANTITATIVE COMPARISON WITH OTHER METHODS ON SYNTHETIC IMAGES

|           | S1      |        | S2      |        | S3      |        | S4      |        | S5      |        |
|-----------|---------|--------|---------|--------|---------|--------|---------|--------|---------|--------|
|           | PSNR    | SSIM   | PSNR    | SSIM   | PSNR    | SSIM   | PSNR    | SSIM   | PSNR    | SSIM   |
| He [15]   | 13.8401 | 0.8189 | 19.6303 | 0.8860 | 13.8780 | 0.6526 | 11.3847 | 0.7010 | 11.5138 | 0.6783 |
| Meng [27] | 20.0077 | 0.8949 | 18.5343 | 0.7760 | 10.7878 | 0.5617 | 17.1678 | 0.7906 | 14.6130 | 0.8262 |
| Zhao [44] | 16.2053 | 0.7001 | 15.3478 | 0.7548 | 16.1868 | 0.4909 | 20.6393 | 0.8289 | 18.1280 | 0.7811 |
| Zhu [46]  | 21.7409 | 0.9346 | 22.0426 | 0.9157 | 19.4366 | 0.8931 | 17.0786 | 0.8851 | 20.7937 | 0.9006 |
| Cai [4]   | 21.3580 | 0.9447 | 21.4769 | 0.9045 | 23.9406 | 0.9283 | 22.3021 | 0.9216 | 23.3178 | 0.9402 |
| Ren [31]  | 23.7623 | 0.9498 | 20.6320 | 0.8334 | 21.0891 | 0.8814 | 22.7745 | 0.8897 | 18.1981 | 0.8742 |
| Engin [8] | 20.9133 | 0.8689 | 19.9102 | 0.8260 | 24.3680 | 0.8932 | 21.3047 | 0.8625 | 17.9281 | 0.8011 |
| Ours      | 24.1396 | 0.9346 | 22.6202 | 0.9242 | 24.6384 | 0.9341 | 23.2002 | 0.9308 | 23.7816 | 0.9305 |

TABLE II  
QUANTITATIVE COMPARISON WITH OTHER METHODS ON REAL WORLD IMAGES

|           | R1     |           |              | R2     |           |              | R3     |           |              | R4     |           |              | R5     |           |              |
|-----------|--------|-----------|--------------|--------|-----------|--------------|--------|-----------|--------------|--------|-----------|--------------|--------|-----------|--------------|
|           | $e$    | $\bar{r}$ | $\delta(\%)$ | $e$    | $\bar{r}$ | $\delta(\%)$ | $e$    | $\bar{r}$ | $\delta(\%)$ | $e$    | $\bar{r}$ | $\delta(\%)$ | $e$    | $\bar{r}$ | $\delta(\%)$ |
| He [15]   | 1.0911 | 1.2249    | 0.2129       | 0.0879 | 1.0959    | 0.0668       | 0.2859 | 1.1270    | 1.3458       | 0.0885 | 1.6350    | 0.1039       | 0.3069 | 1.1718    | 0.0837       |
| Meng [27] | 0.7199 | 1.9378    | 0.7876       | 0.3923 | 1.6495    | 0.1387       | 0.3992 | 1.6197    | 3.8328       | 0.1019 | 2.2125    | 0.2533       | 0.3391 | 1.5334    | 0.1165       |
| Zhao [44] | 0.2342 | 1.6880    | 0.8521       | 0.0354 | 1.8630    | 0.7519       | 0.1618 | 1.3906    | 2.2611       | 0.1504 | 2.1234    | 0.2962       | 0.1634 | 1.7812    | 0.5595       |
| Zhu [46]  | 0.9440 | 1.3380    | 0.8076       | 0.0692 | 1.1446    | 0.0605       | 0.2425 | 0.9892    | 0.7610       | 0.0285 | 1.3958    | 0.2905       | 0.1334 | 0.9497    | 0.4729       |
| Cai [4]   | 0.4131 | 1.4648    | 0.5810       | 0.0052 | 1.2023    | 5.7835       | 0.1859 | 1.0992    | 9.4908       | 0.0405 | 1.4304    | 0.1538       | 0.2152 | 1.1930    | 0.4679       |
| Ren [31]  | 0.4107 | 1.4849    | 0.4538       | 0.0553 | 1.1750    | 1.4064       | 0.1339 | 1.3448    | 2.7439       | 0.0520 | 1.9268    | 0.2347       | 0.1502 | 1.0919    | 0.0757       |
| Engin [8] | 0.0140 | 1.2517    | 0.5293       | 0.0168 | 1.2009    | 0.7820       | 0.1096 | 1.1858    | 3.0451       | 0.0807 | 1.2683    | 0.2507       | 0.1265 | 1.1966    | 0.6675       |
| Ours      | 0.6749 | 1.5404    | 0.4872       | 0.4761 | 1.8872    | 0.0610       | 0.3556 | 1.6333    | 1.5247       | 0.1655 | 1.8516    | 0.2076       | 0.3057 | 1.6942    | 0.0842       |

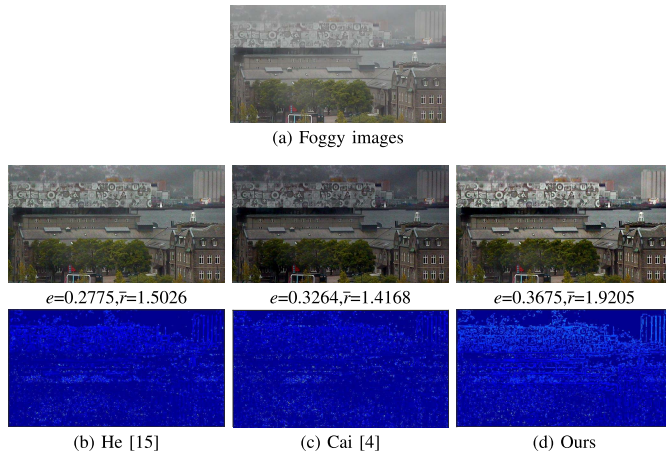


Fig. 14. Comparison of different defogging techniques on real world image of the MRFID. The 2nd row is the defogged results of (a) by different methods. The 3rd row is the visible gradient ratio maps for the corresponding results in 2nd row.

clearer. In addition, we further compare the proposed method with other learning-based methods [38], [42]. As shown in Fig. 11 and Fig. 12, we can see that our results have a better recovered quality than others in contrast and sharpness.

3) *MRFID and Other Datasets*: We further evaluate the proposed method against other defog methods [4], [8], [15], [31], [38], [42], [44], [46] on our real foggy dataset MRFID

and other foggy dataset O-HAZE [2], HazeRD [43]. O-HAZE is an outdoor hazy image database with hazy and haze-free pairs, and it has been employed in the dehazing challenge of the NTIRE 2018 CVPR workshop. This dataset consists of 45 outdoor foggy images and their corresponding clear image(ground truth). HazeRD is also an outdoor hazy image database, which contains fifteen real haze-free outdoor scenes. For each scene, there are five corresponding haze images generated by using the scattering theory through physically realistic parameters.

As shown in Fig. 13, we can see that our method is superior to the others in sharpness and brightness. We further discuss this in Fig. 14, the rate of new visible edges  $e$  and the quality value of the contrast restoration  $\bar{r}$  are larger than other methods, which means our method has more competitive results in local contrast and edge information. Moreover, we can see that our method has a stronger visible gradient ratio than others (as shown in the third row of Fig. 14), which is reflected by the indicator  $\bar{r}$ . It is shown that our method is less prone to artifacts, the generated defogged image is more natural and has high contrast. We also quantitatively evaluate our method on MRFID with FADE and BAVE metrics in Table III. As shown, our method has a better defog performance compared with the other methods [4], [8], [15], [27], [31], [38], [42], [44], [46] in this dataset. To further test our method, we performed a comparison of FRIQA and BIQA for each algorithm on database O-HAZE [2] and HazeRD [43].

TABLE III  
AVERAGE BAVE AND FADE OF DEFOGGED RESULTS ON OUR DATASET MRFID (AVERAGED OVER 800 FOGGY IMAGES)

|               | He [15] | Meng [27]     | Zhu [46] | Cai [4] | Ren [31] | Zhao [44] | Zhang [42] | Wang [38] | Engin [8] | Our           |
|---------------|---------|---------------|----------|---------|----------|-----------|------------|-----------|-----------|---------------|
| $e$           | 0.9439  | <b>2.9764</b> | 1.1095   | 0.7863  | 0.9185   | 2.3678    | 1.0596     | 1.8835    | 1.4337    | 2.8910        |
| $\bar{r}$     | 1.1249  | 2.0890        | 0.9638   | 1.0823  | 1.1810   | 2.2730    | 1.8775     | 1.7490    | 2.1668    | <b>2.5828</b> |
| $\delta(\%)$  | 0.3923  | 0.7939        | 3.9306   | 6.8770  | 2.3482   | 1.8182    | 1.5683     | 0.6414    | 5.5362    | <b>0.3455</b> |
| $\mathcal{F}$ | 1.3105  | 1.0346        | 1.2411   | 1.2171  | 1.3713   | 0.9256    | 1.8989     | 1.6770    | 1.7292    | <b>0.8640</b> |

TABLE IV  
AVERAGE PSNR, SSIM, BAVE AND FADE OF DEFOGGED RESULTS ON O-HAZE/HAZERD DATASET [2], [43]

|               | He [15]             | Meng [27]                 | Zhu [46]            | Cai [4]             | Ren [31]                  | Zhao [44]                | Zhang [42]         | Wang [38]           | Engin [8]           | Our                         |
|---------------|---------------------|---------------------------|---------------------|---------------------|---------------------------|--------------------------|--------------------|---------------------|---------------------|-----------------------------|
| $PSNR$        | 15.8255/<br>14.8411 | 16.1449/<br>15.0128       | 15.4608/<br>14.4993 | 15.6237/<br>15.8470 | 17.1734/<br>15.5382       | 13.0865/<br>14.2701      | 10.2637/<br>8.2787 | 14.2692/<br>11.4923 | 12.8572/<br>15.2993 | <b>17.5637/<br/>16.6354</b> |
| $SSIM$        | 0.3918/<br>0.5733   | 0.5057/<br>0.6060         | 0.3750/<br>0.4863   | 0.4333/<br>0.5422   | <b>0.5203/<br/>0.5958</b> | 0.4215/<br>0.5162        | 0.3423/<br>0.3870  | 0.4358/<br>0.5093   | 0.2921/<br>0.3403   | 0.5153/<br><b>0.6232</b>    |
| $e$           | 1.0832/<br>2.3386   | 1.7561/<br>2.4729         | 1.2872/<br>1.8081   | 0.6541/<br>1.7825   | 1.1454/<br>1.2777         | 1.7196/<br><b>3.2209</b> | 1.3565/<br>1.4734  | 1.1628/<br>1.7731   | 1.6071/<br>2.4933   | <b>1.7858/<br/>3.1489</b>   |
| $\bar{r}$     | 1.4995/<br>2.0872   | 2.4457/<br>3.0618         | 1.2254/<br>1.6129   | 1.2812/<br>1.7738   | 1.8643/<br>1.7085         | 2.1408/<br>2.9980        | 2.1811/<br>2.7336  | 1.8304/<br>1.8769   | 1.5976/<br>2.2770   | <b>2.8533/<br/>3.1870</b>   |
| $\delta(\%)$  | 0.2237/<br>0.1655   | <b>0.2016/<br/>0.1443</b> | 0.3755/<br>0.3614   | 0.3481/<br>0.4838   | 0.4586/<br>0.3843         | 1.3231/<br>1.5398        | 0.2531/<br>0.2386  | 0.2334/<br>0.3116   | 1.5042/<br>1.2137   | 0.2216/<br><b>0.1380</b>    |
| $\mathcal{F}$ | 0.7806/<br>1.1230   | 0.5607/<br>0.8937         | 0.8029/<br>1.2860   | 1.0248/<br>1.4279   | 0.7546/<br>1.4063         | 0.5572/<br>0.6345        | 1.2932/<br>1.4834  | 0.7977/<br>1.5758   | 0.8275/<br>1.1352   | <b>0.5428/<br/>0.5744</b>   |



Fig. 15. Effectiveness of the proposed Cycle-Defog2Refog approach. The 1st and 3rd row of (b)-(g) are the defogged results of (a) by various methods. The 2nd and 4th row of (b)-(g) are an enlarged sub-image highlighted in (a) of the corresponding image in 1st and 3rd row. (a) Foggy Inputs with their corresponding enlarged sub-image, the 1st row image named as 'people' and the 3rd row image named as 'garden'. (b) CycleGan without using cycle consistency loss function. (c) Cycle-Defog2Refog without using cycle consistency loss function. (d) CycleGan [45]. (e) Cycle-Defog2Refog without VGG. (f) Cycle-Defog2Refog without E-net. (g) Cycle-Defog2Refog without E-net, but with a similar number of parameters as the proposed network. (h) Cycle-Defog2Refog.

As shown in TABLE IV, the PSNR and SSIM of our results are almost larger than others. Compared to other methods' defogged results, our results have obvious advantages on  $e$ ,  $\bar{r}$ ,

$\delta$  and  $\mathcal{F}$ . These indicate that our method has better defogging performance than other methods [4], [8], [15], [27], [31], [38], [42], [44], [46].



TABLE V  
QUANTITATIVE ANALYSIS OF THE PSNR/SSIM PERFORMANCES OF  
DIFFERENT CONFIGURATIONS OF OUR CYCLE-DEFOG2REFOG  
METHOD IN FIG.15

|        | (b)               | (c)                | (d)                | (e)                | (f)                | (g)               | (h)                |
|--------|-------------------|--------------------|--------------------|--------------------|--------------------|-------------------|--------------------|
| people | 8.2706/<br>0.2229 | 12.0299/<br>0.3425 | 12.6555/<br>0.5496 | 10.7383/<br>0.4221 | 10.1670/<br>0.4338 | 8.7747/<br>0.3095 | 13.2065/<br>0.6682 |
| garden | 7.6774/<br>0.0910 | 12.1461/<br>0.2527 | 14.0462/<br>0.4696 | 13.4583/<br>0.4650 | 12.4258/<br>0.4379 | 8.3138/<br>0.1089 | 14.3676/<br>0.4782 |

## V. ANALYSIS AND DISCUSSION

### A. Effectiveness of Cycle-Defog2Refog Network

In this section, we analyze how the Cycle-Defog2Refog network is more effective than the original CycleGAN in recovering the clear image from the foggy image. We qualitatively and quantitatively compared the defogged results by using CycleGAN and our method with/without E-net and VGG loss function, as shown in Fig. 15 and TABLE V.

In Fig. 15 (b)-(c), the results are generated by both methods only using the generative adversarial loss function. We can see that the CycleGAN's result (as shown in Fig. 15 (b)) has been distorted and the texture details are completely lost. In contrast, as shown in Fig. 15 (c), although the color distortion appears in the defogged results of our method, the texture details are completely preserved. It is due to we have adopted a refog network based on the atmospheric degeneration model to strengthen the mapping function so that we can preserve more texture details for the defogged results. Moreover, Fig. 15 (d) and Fig. 15 (h) show the defogged results of CycleGAN and our algorithm by using full loss function. As shown in the zoom-in regions in Fig. 15 (d), the bicycle sign is blurred (2nd row) and the small roof are covered by fog (4th row). In our results, as shown in Fig. 15 (h), the bicycle sign has clearer edges and there is no fog remained around the small roof. Fig. 15(e) shows the results by using our method without the VGG loss function. Compared to the results of the CycleGan, we can see that our method has better results even without VGG due to the E-net in our defogging framework. However, compared to the results by using our method with VGG loss function (as shown in Fig. 15(h)), the texture information is clearer. Moreover, as shown in Fig. 15 (f), the results generated by our method without using E-net. Compared to the results by using E-net(as shown in Fig. 15(h)), we can see that the enlarged sub-image looks more dim and fuzzy. In particular, to demonstrate the performance gain yielded by E-Net, rather than simply due to increased network parameters, we did an experiment using Cycle-Defog2Refog without E-net, but with a similar number of parameters as the proposed network. We increase one layer in the Defog-Net and two layers in Refog-Net, the results are shown in Fig. 15 (g). As can be seen that these results are much darker than our results and some texture information is lost. It indicates that the E-net is the key to our method and the two stage mapping strategy has a remarkable performance for image defogging. The results of the above discussion are also reflected in TABLE V, compared with other results, our results have the best PSNR and SSIM.

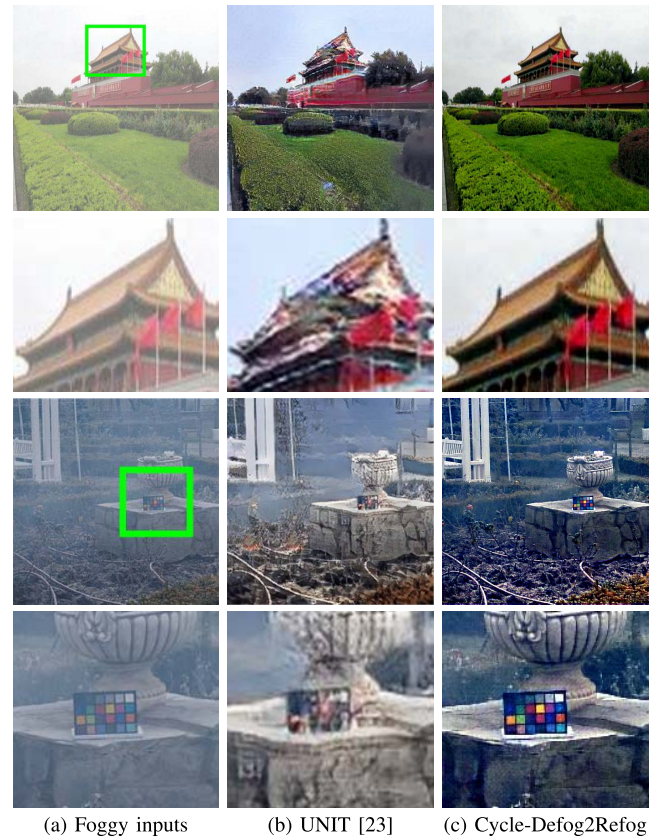


Fig. 16. Comparing the defogged results by using our model and the UNIT model [23]. The 2nd and 4th row of (b)-(c) is an enlarged sub-image highlighted in (a) of the corresponding image in 1st and 3rd row.

To further demonstrate the effectiveness of our method, we compare our model with UNIT model [23]. This model is a CycleGAN based architecture, except that it adds a shared latent space. In addition, the training datasets of both models are RESIDE [21] and O-HAZE [2]. As shown in Fig. 16 (b), the results of the UNIT are distorted (From the enlarged sub-image can be clearly seen.). The reason is that the shared-latent space assumption implies the cycle-consistency constraint. Thus, when the fog obscures the target in the image, the information acquired by the shared space is reduced, resulting in loss of texture feature information during the conversion process. Compared to our method, as shown in Fig. 16 (c), our results are clearer. This further demonstrates that our model has obvious advantages in defogging.

### B. Limitations

As with previous approaches, there are limitations in the proposed approach. One such limitation is that Cycle-Defog2Refog cannot handle heavy fog. As shown in Fig. 17, in the first row, our method as well as those of He [15] and Ren [31] did not handle this image well; In the second row, all the methods have failed to deal with the heavy fog image. The reason is that the current atmospheric degradation model can no longer accurately describe the foggy map in this case. Another limitation is that there are some checkerboard artifacts in our results, such as the brown rooftop of the house in

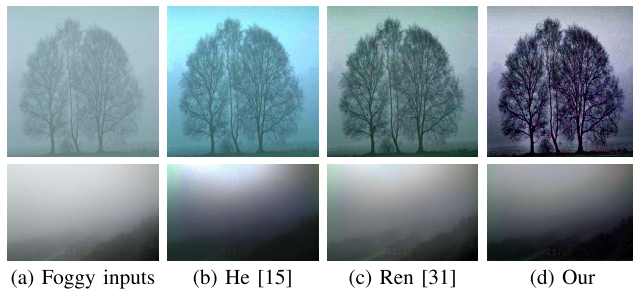


Fig. 17. The proposed approach and the state-of-the-art approaches do not work well for the foggy images of heavy fog density.

the lower middle section of the bottom image in Fig. 11 and around the bag in the bottom image of Fig. 12. The reason for these artifacts may be that “uneven overlap” pattern in deconvolution [11] and the fog has its own particularity to obscure the texture information in the image. Possible solutions to overcome these problems may include a combination of the following, using padding operation before doing convolution in CNNs, training with an appropriate number of iterations and building a multifarious foggy dataset to train our network.

## VI. CONCLUDING REMARKS

In this paper, we adopt a cycle generative adversarial network for single image defogging. Our method is trained by unpaired image data, which avoids preparing a large number of synthetic foggy images in advance. To generate a more real and clearer image, we have proposed a new refog network based on the physical model and a new enhancer network to supervise the mapping from fog domain to fog-free domain. In refog network, we further presented a sky prior to estimating the atmospheric light to prevent artifacts, such as color distortion. Moreover, we introduce a new foggy dataset to train and evaluate our approach, it includes 200 clear outdoor images and 800 foggy images with different fog density. Extensive experiments on both synthetic and real-world foggy images demonstrate that our method performs favorably against several state-of-the-art methods.

## REFERENCES

- [1] C. O. Ancuti, C. Ancuti, C. Hermans, and P. Bekaert, “A fast semi-inverse approach to detect and remove the haze from a single image,” in *Proc. Asian Conf. Comput. Vis.* Berlin, Germany: Springer, 2010, pp. 501–514.
- [2] C. O. Ancuti, C. Ancuti, R. Timofte, and C. De Vleeschouwer, “O-HAZE: A dehazing benchmark with real hazy and haze-free outdoor images,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. NTIRE Workshops (NTIRE CVPR)*, Jun. 2018, pp. 754–762.
- [3] D. Berman, T. Treibitz, and S. Avidan, “Non-local image dehazing,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 1674–1682.
- [4] B. Cai, X. Xu, K. Jia, C. Qing, and D. Tao, “DehazeNet: An end-to-end system for single image haze removal,” *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5187–5198, Nov. 2016.
- [5] B.-H. Chen, S.-C. Huang, and J. H. Ye, “Hazy image restoration by bi-histogram modification,” *ACM Trans. Intell. Syst. Technol.*, vol. 6, no. 4, p. 50, 2015.
- [6] L. Kwon Choi, J. You, and A. C. Bovik, “Referenceless prediction of perceptual fog density and perceptual image defogging,” *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3888–3901, Nov. 2015.
- [7] A. Dudhane and S. Murala, “RYF-net: Deep fusion network for single image haze removal,” *IEEE Trans. Image Process.*, vol. 29, pp. 628–640, 2020.
- [8] D. Engin, A. Genç, and H. K. Ekenel, “Cycle-dehaze: Enhanced CycleGAN for single image dehazing,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2018, pp. 825–833.
- [9] R. Fattal, “Dehazing using color-lines,” *ACM Trans. Graph.*, vol. 34, no. 1, pp. 13:1–13:14, Dec. 2014.
- [10] M. Abadi *et al.*, “TensorFlow: Large-scale machine learning on heterogeneous distributed systems,” 2016, *arXiv:1603.04467*. [Online]. Available: <https://arxiv.org/abs/1603.04467>
- [11] C. Godard, O. M. Aodha, and G. J. Brostow, “Unsupervised monocular depth estimation with left-right consistency,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 270–279.
- [12] I. Goodfellow *et al.*, “Generative adversarial nets,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
- [13] N. Hautière, J.-P. Tarel, D. Aubert, and É. Dumont, “Blind contrast enhancement assessment by gradient ratioing at visible edges,” *Image Anal. Stereol. J.*, vol. 27, no. 2, pp. 87–95, Jun. 2008.
- [14] K. He, J. Sun, and X. Tang, “Fast matting using large kernel matting Laplacian matrices,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2010, pp. 2165–2172.
- [15] K. He, J. Sun, and X. Tang, “Single image haze removal using dark channel prior,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011.
- [16] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
- [17] N. Jacobs, N. Roman, and R. Pless, “Consistent temporal variations in many outdoor scenes,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2007, pp. 1–6.
- [18] J. Johnson, A. Alahi, and L. Fei-Fei, “Perceptual losses for real-time style transfer and super-resolution,” in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2016, pp. 694–711.
- [19] A. Levin, D. Lischinski, and Y. Weiss, “A closed-form solution to natural image matting,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 2, pp. 228–242, Feb. 2008.
- [20] B. Li, X. Peng, Z. Wang, J. Xu, and D. Feng, “AOD-net: All-in-one dehazing network,” in *Proc. IEEE Int. Conf. Comput. Vis.*, vol. 1, Oct. 2017, p. 7.
- [21] B. Li *et al.*, “Benchmarking single image dehazing and beyond,” 2017, *arXiv:1712.04143*. [Online]. Available: <http://arxiv.org/abs/1712.04143>
- [22] R. Li, J. Pan, Z. Li, and J. Tang, “Single image dehazing via conditional generative adversarial network,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, vol. 3, Mar. 2018, p. 24.
- [23] M.-Y. Liu, T. Breuel, and J. Kautz, “Unsupervised image-to-image translation networks,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 700–708.
- [24] P.-J. Liu, S.-J. Horng, J.-S. Lin, and T. Li, “Contrast in haze removal: Configurable contrast enhancement model based on dark channel prior,” *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2212–2227, May 2019.
- [25] J. Mai, Q. Zhu, D. Wu, Y. Xie, and L. Wang, “Back propagation neural network dehazing,” in *Proc. IEEE Int. Conf. Robot. Biomimetics (ROBIO)*, Dec. 2014, pp. 1433–1438.
- [26] E. J. McCartney, *Optics of the Atmosphere: Scattering By Molecules and Particles*. New York, NY, USA: Wiley, 1976, p. 421.
- [27] G. Meng, Y. Wang, J. Duan, S. Xiang, and C. Pan, “Efficient image dehazing with boundary constraint and contextual regularization,” in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 617–624.
- [28] A. Mittal, A. K. Moorthy, and A. C. Bovik, “No-reference image quality assessment in the spatial domain,” *IEEE Trans. Image Process.*, vol. 21, no. 12, pp. 4695–4708, Dec. 2012.
- [29] S. K. Nayar and S. G. Narasimhan, “Vision in bad weather,” in *Proc. 7th IEEE Int. Conf. Comput. Vis.*, Jun. 1999, pp. 820–827.
- [30] N. Otsu, “A threshold selection method from gray-level histograms,” *IEEE Trans. Syst., Man, Cybern.*, vol. 9, no. 1, pp. 62–66, Jan. 1979.
- [31] W. Ren, S. Liu, H. Zhang, J. Pan, X. Cao, and M.-H. Yang, “Single image dehazing via multi-scale convolutional neural networks,” in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2016, pp. 154–169.
- [32] W. Ren *et al.*, “Gated fusion network for single image dehazing,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3253–3261.
- [33] N. Silberman, D. Hoiem, P. Kohli, and R. Fergus, “Indoor segmentation and support inference from RGBD images,” in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2012, pp. 746–760.



- [34] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015, pp. 1–14.
- [35] J. Sun, J. Jia, C.-K. Tang, and H.-Y. Shum, "Poisson matting," *ACM Trans. Graph.*, vol. 23, no. 3, pp. 315–321, Aug. 2004.
- [36] R. T. Tan, "Visibility in bad weather from a single image," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2008, pp. 1–8.
- [37] K. Tang, J. Yang, and J. Wang, "Investigating haze-relevant features in a learning framework for image dehazing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2995–3000.
- [38] A. Wang, W. Wang, J. Liu, and N. Gu, "AIPNet: Image-to-image single image dehazing with atmospheric illumination prior," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 381–393, Jan. 2019.
- [39] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [40] D. Yang and J. Sun, "Proximal dehaze-net: A prior learning-based deep network for single image dehazing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 702–717.
- [41] H. Zhang and V. M. Patel, "Densely connected pyramid dehazing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 3194–3203.
- [42] H. Zhang, V. Sindagi, and V. M. Patel, "Multi-scale single image dehazing using perceptual pyramid deep network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2018, pp. 902–911.
- [43] Y. Zhang, L. Ding, and G. Sharma, "HazeRD: An outdoor scene dataset and benchmark for single image dehazing," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3205–3209.
- [44] D. Zhao, L. Xu, Y. Yan, J. Chen, and L.-Y. Duan, "Multi-scale optimal fusion model for single image dehazing," *Signal Process., Image Commun.*, vol. 74, pp. 253–265, May 2019.
- [45] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.
- [46] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.



**Wei Liu** received the M.Eng. degree in control science and engineering and the Ph.D. degree in computer science and technology from the Wuhan University of Technology in 2013 and 2017, respectively. He has been a Postdoctoral Researcher with Shenzhen University, China, since 2018. His research interests include computer vision, deep learning, and image processing.



**Xianxu Hou** received the B.S. and M.S. degrees from the China University of Mining and Technology at Beijing in 2011 and 2014, respectively, and the Ph.D. degree in computer science from the University of Nottingham in 2018. He is currently a Postdoctoral Researcher with the College of Information Engineering, Shenzhen University, China. His research interests include deep learning, computer vision, and natural language processing.



**Jiang Duan** received the B.S. degree in mechanical engineering from Southwest Jiaotong University, the M.S. degree from the University of Derby in 2002, and the Ph.D. degree from the University of Nottingham in 2006. He is currently a Professor with the Southwestern University of Finance and Economics, China. He is also an Expert of the National Thousand Talents Plan, the Winner of the first Sichuan Outstanding Talent Award (the highest talent award of Sichuan province), the Winner of the Sichuan Youth Science and Technology Award, the Standing Committee Member of Sichuan Association for Science and Technology, and the Vice Chairman of Sichuan Science and Youth Federation. He has published more than 30 academic articles, won more than ten international and national patents, and his researches have been funded by about 20 national and provincial funds.



**Guoping Qiu** is currently a Distinguished Professor of information engineering, the Director of Shenzhen University Intelligent Robotics Centre, Shenzhen University, China, and a Chair Professor of visual information processing with the University of Nottingham, Nottingham, U.K. He has taught in universities in the U.K. and Hong Kong and also consulted for multinational companies in Europe, Hong Kong, and China. His research interests include image processing, pattern recognition, and machine learning. He is particularly known for his pioneering research in high-dynamic range imaging and machine learning-based image processing technologies. He has published widely and holds several European and U.S. patents. Technologies developed in his lab have laid the cornerstone for successful spinout companies who are developing advanced digital photography software enjoyed by tens of millions of global users.